Worksheet No. 9

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Subject Name: Technical Training **Subject Code:** 25CAP-652

1.Aim/Overview of the practical:

To understand and implement triggers in PostgreSQL.

2. Objectives:

To understand and implement triggers in PostgreSQL.

3. Input/Apparatus Used:

- PostgreSQL
- pgAdmin

4.Procedure/Algorithm/Code:

1. Create the table:

```
CREATE TABLE employee (
emp_id INT PRIMARY KEY,
emp_name VARCHAR (50),
working_hours INT,
perhour_salary NUMERIC,
total_payable_amount NUMERIC
);
```

```
CREATE TABLE

Query returned successfully in 153 msec.
```

2. Creating a function

```
CREATE OR REPLACE FUNCTION CACULATE_PAYABLE_AMOUNT () RETURNS
TRIGGER
AS
$$
 BEGIN
            NEW.total_payable_amount:=NEW.perhour_salary*New.working_hours;
```

```
        IF NEW.total_payable_amount>25000 THEN
        RAISE EXCEPTION 'INVALID ENTRY BECAUSE PAYABLE AMOUNT CAN NOT BE GREATER THAN 25000';

        END IF;

        RETURN NEW;
  END;

$$ LANGUAGE PLPGSQL;
```

Data Output | Messages | Notifications

```
CREATE FUNCTION

Query returned successfully in 698 msec.
```

3. Create a trigger

```
CREATE OR REPLACE TRIGGER
AUTOMATED_PAYABLE_AMOUNT_CALCULATION
BEFORE INSERT
ON  employee
FOR EACH ROW
EXECUTE FUNCTION CACULATE_PAYABLE_AMOUNT()
```

```
CREATE TRIGGER

Query returned successfully in 114 msec.
```

```
INSERT INTO employee (emp_id, emp_name, working_hours, perhour_salary)
VALUES (1, 'AKASH', 10, 1000);
```

```
INSERT 0 1

Query returned successfully in 105 msec.
```

5. Learning Outcomes:

1. Using TCL commands COMMIT, ROLLBACK and SAVEPOINT in PostgreSQL.
2. Transactions in DBMS.